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Classification of plastics based on laser-induced breakdown spectroscopy using one-shot learning in conditional generative adversarial networks

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ABSTRACT

Facing increasing plastic use year by year, there is serious plastic pollution around the world. In the application of plastic detection and classification using laser-induced breakdown spectroscopy (LIBS) spectra, it has become evident that classification models highly depend on the amount of data. However, in complex industrial environments, the quality and quantity of LIBS spectra cannot be effectively guaranteed. Therefore, it becomes critical that the accuracy of plastic classification under few-shot or even one-shot learning be improved. In this work, we proposed a data augmentation method based on LIBS. A modified generative adversarial network (GAN), called Conditional GAN (C-GAN), was developed to expand the data of single experimental plastic spectra. Then BP, k-NN and SVM networks were used to demonstrate the accuracy improvement of plastic classification. The results of BP network showed that the validation accuracy of one-shot plastic classification increased from 90.44% to 99.49%. The k-NN and SVM also provided similar results. In addition, compared to the experimentally obtained two-shots plastic classification, the classification accuracy increased from 98.95% to 99.87% when the training set containing two experimental spectra was augmented with synthetic spectra generated by C-GAN, and the stability was also improved. In a word, this work provides a feasible method to increase the validation accuracy of one-shot plastic classification, thereby facilitating the application of LIBS in industrial plastic classification fields.

1. Introduction

With light weight, stable chemical properties, strong impact resistance and simple production, plastic is widely used with about 420 million tons in 2024 worldwide. Plenty of discarded plastic debris accumulates in various natural habitats, affecting numerous species[1]. Therefore, the classification and recycling of plastic waste have always been a focal point in environmental governance efforts. Conventional methods for plastic classification include Differential Scanning Calorimetry (DSC), infrared spectroscopy, UV–VIS absorption spectroscopy, Raman spectroscopy and so on. However, DSC[2] and Fourier Transform Infrared (FTIR)[3,4] cannot work in-line, and they require complex and diverse pretreatment as well as lengthy analytical time. Absorption bands in the UV–Vis range are often broad and overlapping, making

their interpretations not always straightforward[5]. Raman spectroscopy is easily affected by strong fluorescence, which is simultaneously excited with the Raman signal[6,7]. Most of these methods require stringent operating conditions. Thus, they are difficult to apply on a large scale in industrial fields. Laser-induced breakdown spectroscopy (LIBS) technology is a promising method with advantages of simultaneous analysis of multiple elements, remote sensing, rapid response and no need for preparation. The spectrum in LIBS contains abundant atomic and molecular information in plastics[8]. Thus, LIBS is more suitable for online identifying and sorting discarded plastic debris in industrial fields [9–11].

More deep learning models[12–14] were generally used for fast feature recognition in LIBS spectra for plastic classification. However, the traditional deep learning models rely on many labeled data to realize

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high accuracy. For too many types of plastics in the recycling industries, it is difficult to obtain such massive data with labels. With a small number of samples for model training, achieving high accuracy and efficiency has become a challenge. In order to address the challenge of LIBS quantification with small samples in Mars exploration, Cui et al. [15] proposed a new scheme by integrating a transfer learning technique. They used LIBS data acquired by ChemCam as the target domain and pre-trained the model with a large number of spectra obtained in the laboratory (source domain), followed by knowledge transfer. This improved model performance under small-sample conditions. Choi et al. [16] introduced an unsupervised method, through utilizing principal component analysis (PCA) for dimensionality reduction followed by k-means clustering to classify the grade and origin of iron ore, this achieved a classification accuracy of over 94% without labeled data, demonstrating the potential of unsupervised techniques in scenarios where labeled samples are scarce. While alternative approaches such as unsupervised learning and transfer learning can alleviate the problem of scarce labeled samples to some extent, they face inherent limitations: unsupervised methods struggle with fine-grained discrimination, and transfer learning requires a substantial amount of source domain data.

Data augmentation can expand training data by modifying existing samples through transformations, enhancing model generalization. Generative adversarial net (GAN) is a model commonly used for data augmentation. It was first proposed by Ian J. Goodfellow et al. as a generative model for synthesizing images in computer science [17]. Conventionally, GAN consists of two basic networks: a generator G and a discriminator D . The G model is used to generate data while the D model is used to discriminate the authenticity of the synthetic data and provide feedback about the difference. The loss function is shared by the G model and the D model. When the objective of the G model is to make the distribution of the synthetic data close to the original data – minimizing the loss function, the D model maximizes the loss function in order to distinguish the difference of distribution between synthetic data and original data. Thus, the adversarial effect is achieved via a minimax game engaged in by both networks sharing the loss function. Teng et al. [18] introduced GAN for generating LIBS spectra and combining them with the experimental spectra to build a classification model. Ren et al. [19] designed a Generative Adversarial Neural Network (GANN) with physical loss constraints to augment the LIBS spectral database, and improved the accuracy of machine learning models in the quantitative analysis of Ni, Co, and Li in deep-sea polymetallic nodules. In the research of Jamali et al. [20], a novel data augmentation framework based on transformer conditional Wasserstein generative adversarial network with gradient penalty (TCWGAN-GP) is introduced to selectively generate synthetic LIBS spectra for underrepresented classes in the training set. Yu et al. [21] used Raman spectroscopy combined with GAN to classify three Marine pathogens, which not only improved the accuracy of machine learning classification but also solved the problem requiring a large amount of data. Du et al. [22] proposed a method for classifying three food-borne pathogens. This method was based on Raman spectroscopy combined with a generative adversarial network (GAN) and a multi-class support vector machine (SVM). Using 19 feature variables, the overall classification accuracy was improved to 90%. By combining optical tweezers Raman spectroscopy with Progressive Growing of GAN (PGGAN) and Residual network (ResNet), Liu et al. [23] improved the classification accuracy of 5 deep-sea strains, and reduced the spectral collection time to 1/3. Di Frischia et al. [24] proposed a pipeline for augmenting data with a GAN reinforcement, created a dataset large enough for training three types of neural networks (binary-NN, categorical-NN and CNN). Xu et al. [25] designed a conditional table GAN (CTGAN) to address the challenges in table data. In such data, continuous columns may exhibit multiple modes, while discrete columns are sometimes imbalanced. Therefore, assisted by GAN model for the data augmentation, small sample classification models effectively improve accuracy and efficiency, significantly reducing requirement of data amount.

There are many types of traditional GANs, but many of them suffer from issues such as training instability. For instance, while DCGAN [26] introduced the CNN architecture into GANs and significantly improved the quality of generated images, it still faces the problem of unstable training. On the other hand, Wasserstein GAN (WGAN) [27] makes progress toward stable training of GANs, but sometimes can still generate only poor samples or fail to converge. CTGAN achieves controllable generation by incorporating conditional variables and improves training stability by using gradient penalty instead of weight clipping [28]. However, given only one sample as input, CTGAN cannot be trained or generate data. This is due to its reliance on the Variational Gaussian Mixture Model (VGM) for global statistical analysis of the training set and the packing mechanism of the PacGAN framework. Similarly, traditional GANs suffer from mode collapse or overfitting under data constraints, especially when trained on a single sample [29]. Generator mode collapse would cause the network to output meaningless noise or duplicate the input sample, and overfitting would cause the discriminator to directly memorize the sample features without learning generalization rules. As a result, GAN would cause the discriminator to degenerate into a binary discriminator, and the generator to lose gradient guidance. Therefore, the spectral data collected in industrial settings are characterized by high dimensionality and significant fluctuations in signal-to-noise ratio. Even when enough samples are collected, the problem of mode collapse is highly prone to occur when the training of traditional GAN is inherently unstable. Meanwhile, in plastic recycling production lines, certain special categories of plastics, such as new materials, may be represented by only a single or a very small number of samples. Under such circumstances, conditional generation methods based on global statistics, such as CTGAN, are rendered completely ineffective, and augmented data for these categories cannot be generated. As a result, rare categories remain unable to be provided with sufficient effective training samples.

To address the issues of poor performance in one-shot classification and the inability of traditional GAN to generate data from a single spectral input, in this paper, we proposed a new approach of Conditional GAN (C-GAN) for plastic classification with one-shot learning, overcoming the limitations of the conventional GAN model. Before inputting to C-GAN, the spectral data was fused with relative noise perturbations, spectral line shifts, and nonlinear intensity scaling, to simulate experimental perturbations. Conditional constraints applied to the synthetic spectra ensured similarity to the experimental spectra. The sole experimental spectrum for model training was extended by C-GAN. The synthetic spectrum output from by C-GAN and the sole experimental spectrum were input into a Back Propagation (BP) neural network for training. The classification performance of the models before and after data augmentation were evaluated. The purpose of this paper is to address the issue of large data requirements and improve the model's one-shot classification capability.

2. Experiment

The LIBS system in this work was illustrated in Fig. 1. A Q-switched Nd:YAG laser with the pulse repetition rate of 10 Hz, single pulse energy of 40 mJ, pulse duration of 6 ns, wavelength of 532 nm was used for plasma generation. The laser beam was reflected by a dichroic mirror and focused vertically through a lens ($f=150$ mm) onto the plastic surface. The plastic was moved by a three-dimensionally motorized stage. The plasma optical emission was coupled to an optical fiber, then transmitted to a mechele spectrometer (Andor Mechelle 5000, $\lambda/\Delta\lambda = 5000$, range 200–980 nm) equipped with an ICCD (Andor iStar 334 T). The optical signal was recorded after a 2.5 μ s delay after laser ablation and integrated for 2 μ s. Each spectrum was accumulated by 10 laser shots. The spectral data were transmitted to computer and processed by the proposed classification algorithm.

The plastics used in this work (Table S1 in Supplementary Material) were industrial plastics purchased from 20 manufacturers. The number

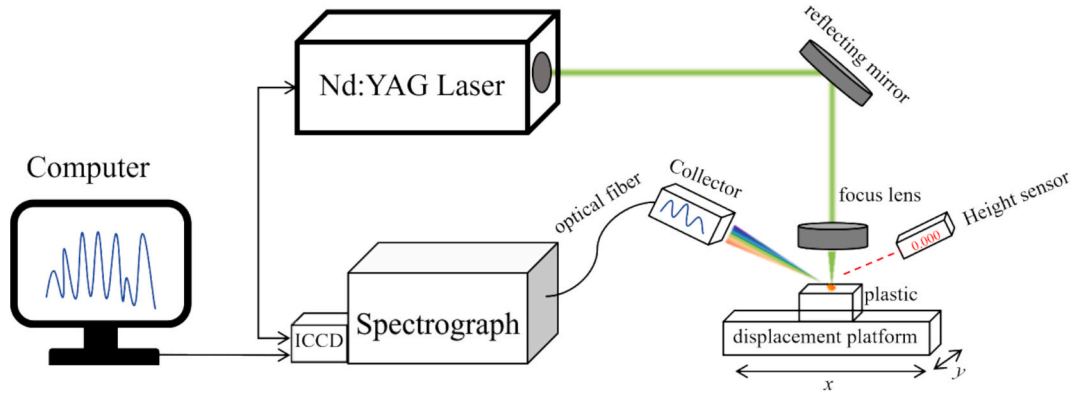


Fig. 1. Schematic diagram of the experimental setup.

of spectra for each kind of plastic was 100. These spectra were collected under consistent conditions, with no batch-to-batch variation.

3. Methodology

Conventional CTGAN can effectively fit and generate continuous columns by using methods such as VGM, Wasserstein-GP loss, the PacGAN framework and so on. But as a kind of deep learning model, it's still limited by the constraint of the number of training samples and are unable to generate high-reliable synthetic spectra in a single sample. The conditional loss and packing mechanism cannot handle one-shot sample.

Therefore, a Conditional GAN (C-GAN) with excellent fitting ability for continuous columns serves as a modified version in this work for breaking the limitation of CTGAN. The model specifically enhanced for handling single spectrum (continuous variable) inputs by introducing a hybrid augmentation strategy and refining the conditional loss, while retaining the architecture suitable for continuous column generation. The structure diagram of C-GAN is shown in Fig. 2.

The structural and parameter modifications are outlined below:

I. Before feeding data into the model: Data containing both continuous and discrete mixed features can be handled by the net. When only continuous data such as spectra was input, the discrete columns parameters in the data sampler should be set to an empty list.

II. Data preprocessing: Non-Gaussian and multimodal distributions are fitted and estimated using the Variational Gaussian mixture Model (VGM) prior to continuous column generation, through which they are normalized and transformed into standard normal distributions. Let c_{ij} be a continuous value corresponding to the i th continuous column C_i and the j th row. The probabilistic model $P_{C_i}(c_{ij})$ was produced by VGM,

a Gaussian Mixture of m_i components with means $\eta^{(1)}, \dots, \eta^{(m_i)}$, standard deviations $\phi_i^{(1)}, \dots, \phi_i^{(m_i)}$ and weights $u^{(1)}, \dots, u^{(m_i)}$ respectively.

$$P_{C_i}(c_{ij}) = \sum_{k=1}^{m_i} u^{(k)} N(c_{ij}; \eta_i^{(k)}, \phi_i^{(k)}) \quad (1)$$

III. Discriminator: The essence of the discriminator is a conditional Wasserstein discriminator. The fully connected networks, the PacGAN framework, and Wasserstein-GP loss function are utilized in the discriminator, which avoids gradient vanishing or explosion when continuous columns are processed, and enables continuous variables with complex distributions, including Gaussian mixtures, to be better learned.

The discriminator was composed of multiple fully connected layers. Both the input and output layers were linear layers. Linear transformations, LeakyReLU activation functions (with a negative slope of 0.2), and Dropout were sequentially applied by the hidden layers.

Data packed under the PacGAN strategy was received by the first linear layer of the discriminator, where the packed data consisted of pac samples and their corresponding conditional vectors, concatenated in the same order. In contrast to CTGAN, where conditional vectors were derived from discrete columns, the conditional vectors in C-GAN were refined to comprise statistical features of the spectrum, addressing the specificity of having only continuous columns. Additionally, the packaging mechanism facilitates the learning of inter-sample correlations and statistical features through simultaneous comparison of distributions across multiple continuous variables, matching the generator structure and ensuring training balance. The number of samples is as many as the batch size m . Then the samples which come from sampling the training data $real_j$ (for $1 \leq j \leq m$), the output of the conditional

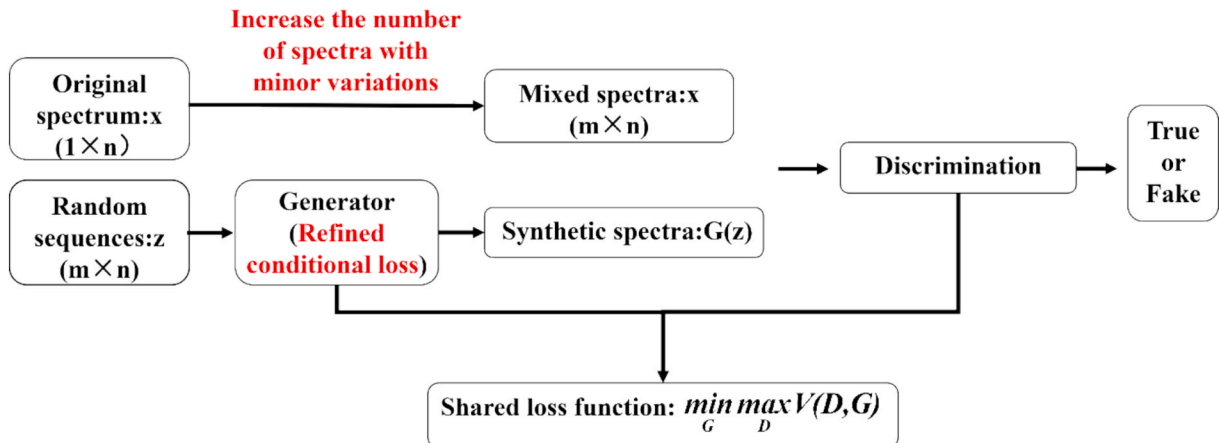


Fig. 2. The structure diagram of C-GAN.

generator $fake_j$ and their corresponding conditional vectors $C_j^{(real)}, C_j^{(fake)}$ are processed to pass 10 “packed” or concatenated them to the discriminator, and detailed like:

$$real_k^{(pac)} = (real_{k \times pac+1} \oplus C_{k \times pac+1}^{(real)}) \oplus \dots \oplus (real_{k \times pac+pac} \oplus C_{k \times pac+pac}^{(real)}) \quad (2)$$

$$fake_k^{(pac)} = (fake_{k \times pac+1} \oplus C_{k \times pac+1}^{(fake)}) \oplus \dots \oplus (fake_{k \times pac+pac} \oplus C_{k \times pac+pac}^{(fake)}) \quad (3)$$

where pac is the number of each package, and $1 \leq k \leq m/pac$.

before fed-forwarded into the D model to obtain the loss of discriminator L_D , to enforce the Lipschitz constraint on the discriminator, ensuring that its gradient norm remains close to 1 in the region between real and generated samples, interpolation samples $\tilde{r}_k^{(pac)}$ are obtained by linearly mixing real $real_k^{(pac)}$ and fake samples $fake_k^{(pac)}$ with random weights $\rho_k^{(pac)}$. These interpolation samples are then used to compute the gradient norm in the gradient penalty, and it follows (6) to compute the gradient penalty L_{GP} for the discriminator.

$$L_D = \frac{1}{m/pac} \sum_{k=1}^{m/pac} D(fake_k^{(pac)}) - \frac{1}{m/pac} \sum_{k=1}^{m/pac} D(real_k^{(pac)}) + 10L_{GP} \quad (4)$$

$$\tilde{r}_k^{(pac)} = \rho_k^{(pac)} fake_k^{(pac)} - (1 - \rho_k^{(pac)}) real_k^{(pac)} \quad (5)$$

$$L_{GP} = \frac{1}{m/pac} \sum_{k=1}^{m/pac} (\|\nabla_{\tilde{r}_k^{(pac)}} D(\tilde{r}_k^{(pac)})\|_2 - 1)^2 \quad (6)$$

where $\rho_1, \dots, \rho_{m/pac} \sim \text{uniform}(0,1)$. Discriminator training steps shown in Fig. 3.

However, when only a single sample is input, the Pac discriminator

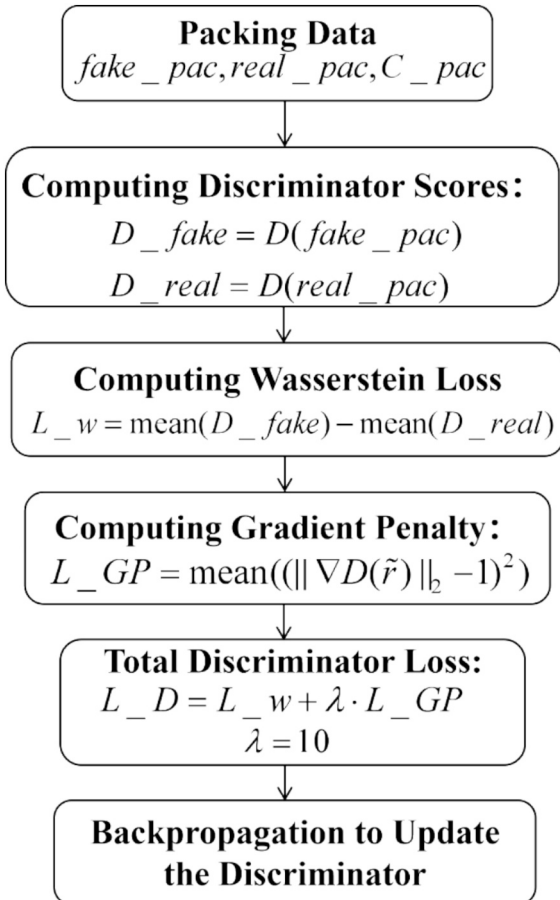


Fig. 3. Training steps of Discriminator.

switches from batch to individual sample discrimination, which prevents the detection of statistical correlations among samples, compromises training stability, and increases the risk of mode collapse.

In this work, a hybrid augmentation strategy for spectral data was proposed. Based on the original individual spectral data, experimental disturbances were simulated by adding relative noise perturbations, spectral line shifts, and nonlinear intensity scaling to increase the number of spectra with minor variations. For noise addition, in order to adapt to the original spectrum, 2% of its standard deviation was used as the noise standard deviation if the standard deviation was non-zero; otherwise, a fixed value of 2 was directly applied. Gaussian white noise was then employed to add independent random noise to each data point within the same spectrum, with noise independently generated for each of the ten distinct augmented versions. Spectral lines were randomly shifted within a small range of $[-0.1 \text{ nm}, 0.1 \text{ nm}]$ to simulate instrumental perturbations. Concurrently, nonlinear intensity scaling was performed within a range of $[0.8, 1.2]$, introducing varied yet subtle intensity changes across spectral points to enhance the robustness of the spectral data.

IV. Generator: Shared residual blocks were adopted by the generator, the skip connections facilitate the retention of more details from the original input features during continuous column generation and alleviate gradient issues in the training of networks. The last residual block connected to the output layer. The output layer was a linear layer, and it was followed by the Tanh and Gumbel Softmax activation functions to map the data to the transformed dimension.

But discrete columns were specified as condition vectors and their conditional loss was calculated in the generator to constrain the generation of continuous data with specific properties in the original model. Since the main purpose of this study is to generate continuous variables with similar distributions but not completely identical. Therefore, the constraints can be changed to the mean of the spectrum, peaks intensity and other statistical features[19], to ensure the spectral similarity of the output data.

$$L_G = -\frac{1}{m/pac} \sum_{k=1}^{m/pac} D(fake_k^{(pac)}) + L_{cond} \quad (7)$$

The original conditional loss function was implemented by calculating the cross-entropy loss of specified discrete columns. Since spectral data exclusively comprise continuous columns, L_{cond} has been modified to compute the mean squared error and statistical properties (mean and variance) of the spectrum. The loss of mean squared error L_{MSE} measures the discrepancy between the generated and real spectrum, making sure that the generated spectrum matches the global intensity distribution of the real spectrum. The loss of mean and variance L_{M-V} ensures that these statistical properties of the generated spectra are consistent with those of the real spectra. The conditional loss function is defined as follows (8):

$$L_{cond} = \lambda_{MSE} L_{MSE} + \lambda_{M-V} L_{M-V} \quad (8)$$

where λ represents the weighting coefficients that balance the contributions of different loss terms, and their values are set as follows based on experimental experience: $\lambda_{MSE} = 0.015$, $\lambda_{M-V} = 0.65$. Generator training steps shown in Fig. 4.

4. Result and discussion

4.1. Wavelet peak searching

A large amount of data points in the LIBS spectrum made little contribution to classification, while only peaks of atomic and molecular lines made a greater contribution. For the LIBS time-frequency signal with a large number of unstable components like migration or mutation, the continuous wavelet was used to extract the time-frequency information of signals in localization[30], for improving classification

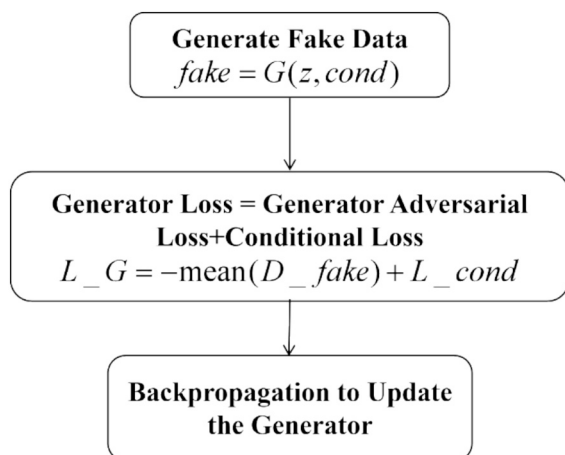


Fig. 4. Training steps of Generator.

efficiency. The noise of the signal was eliminated by level-dependent adaptive soft thresholding[31] and the peaks were selected from the two-dimensional wavelet coefficient matrix[32]. Then, by controlling the threshold of signal-to-noise ratio (SNR), the spectral peak was effectively searched and filtered from more than 20,000 points to approximately 1000 peak points. Fig. 5 shows the comparison of spectrum before and after noise reduction and peak searching.

4.2. Evaluation of data augmentation

On each kind of plastics, 100 spectra were acquired and divided into two datasets for training and testing. The division strategy for the training and test sets in the classification model was as follows: all 100 spectra were first randomly shuffled and then divided as a whole. For example, if the training set required 10 spectra, the first 10 spectra after random shuffling were used as the training set, and the remaining 90 were used as the test set. This ensured that the model had not seen the test set spectra during training. For the plastic classification learning, the number of input layer nodes in the BP neural network was set to 1000 as the dimension of LIBS spectra after peak searching. The sigmoid function served as the activation function, with 35 hidden layer nodes and 20 output layer nodes (the number of plastic types). The goal error and learning rate of the neural network training were set to 0.01 and 0.1, respectively. The classification accuracy of the neural network depends on the number of the training sample, shown in Fig. 6. One-shot learning resulted in poor classification accuracy and stability. The results show that classification accuracy greatly depends on amount of data for training, but it cannot be guaranteed in harsh outdoor industrial environments. Therefore, if the data for training can be augmented, it is possible for improving plastic classification.

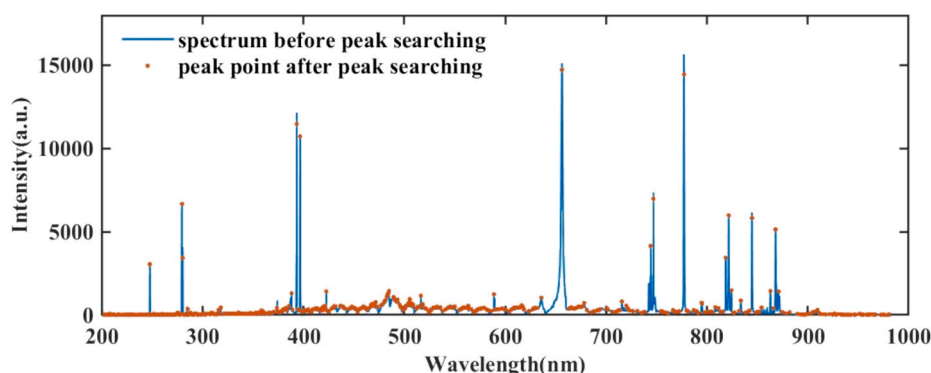


Fig. 5. The spectrum before and after wavelet peak searching.

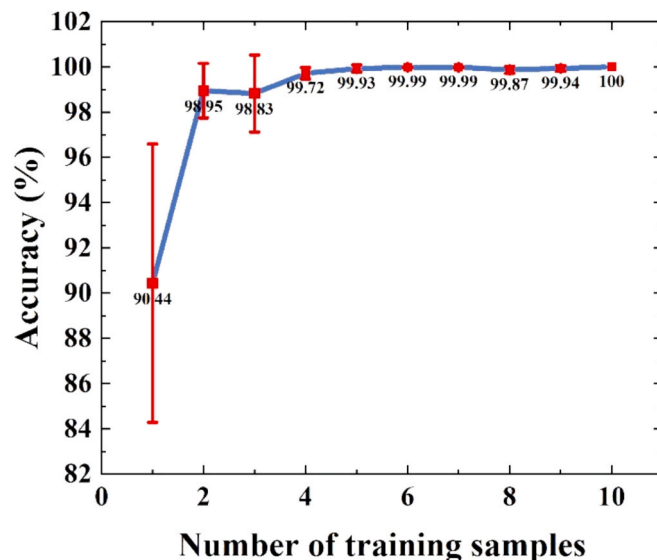


Fig. 6. The classification result of different (1–10) training sample numbers (The remaining samples of 100 spectra were used as the test set).

Before being fed to C-GAN, the spectral data was implemented with noise perturbation, spectral line shift, and nonlinear intensity scaling. The synthetic spectra from different training iterations were evaluated for similarity with experimental spectra using multiple metrics: the difference in average intensity, difference in sample standard deviation, histogram similarity, Manhattan distance, and Pearson correlation coefficient. As shown in Table 1, the synthetic spectra achieved high similarity levels when the number of training iterations reached approximately 500.

For typical examples, the synthetic spectra of ABS1, ABS2, PA1 and PA2 and their corresponding experimental spectra were shown in Fig. 7. Comparing the synthetic and experimental spectra, it can be observed that the synthetic spectra exhibit complete elemental composition and accurate spectral line distribution. High similarity of synthetic and experimental spectra was obtained under 500 interactive times (Table 2). The sample standard deviation between the synthetic spectrum and the experimental spectrum was not significantly different, and the distribution was largely consistent, with a similarity of approximately 0.95. For plastics, the introduction of additives can regulate material properties. However, additives such as flame retardants, pigments, and stabilizers introduce additional atomic spectral lines, which interfere with the classification and identification of plastics. Considering the four types of plastics in Fig. 7, (a1) and (b1) are both ABS, while (c) and (d) are both PA. Nevertheless, the difference in the intensity of the Ca II spectral lines originating from the additives may cause ABS1 and PA1 to be considered more similar, making them appear

Table 1
Similarity between synthetic and experimental spectra at different training iterations.

training iterations	Difference in Average Intensity	Difference in Sample Standard Deviation	Histogram Intersection Similarity	Manhattan distance	Pearson correlation coefficient
1	16.865	24.022	0.929	0.729	0.987
100	30.285	39.469	0.901	0.718	0.986
200	41.921	38.183	0.896	0.724	0.986
300	24.275	20.487	0.918	0.725	0.987
400	−13.408	23.640	0.922	0.702	0.986
500	11.498	15.622	0.948	0.765	0.992
600	24.182	37.166	0.932	0.722	0.987
700	13.366	26.543	0.937	0.732	0.988
800	−31.096	5.281	0.925	0.731	0.990
900	−15.430	15.498	0.935	0.719	0.988
1000	23.215	26.056	0.914	0.702	0.983

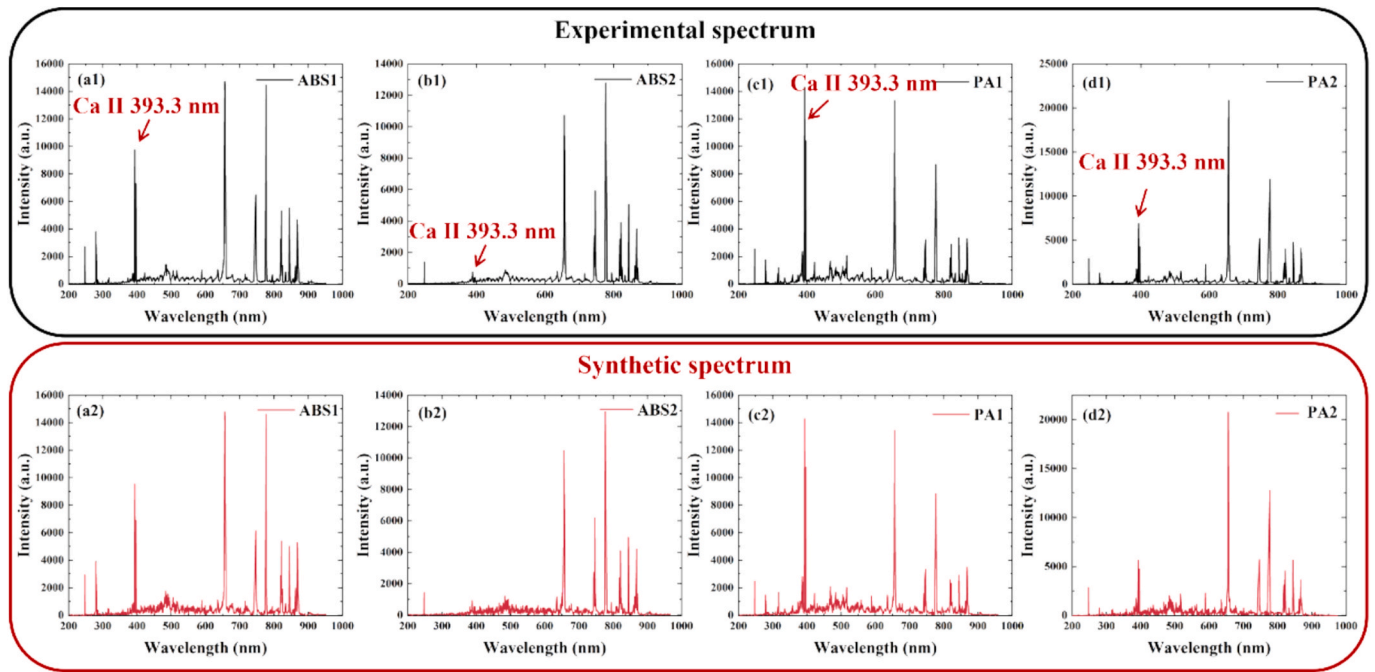


Fig. 7. The synthetic spectrum and the experimental spectrum of ABS1, ABS2, PA1 and PA2.

Table 2
Similarity between synthetic and experimental spectrum at ABS1, ABS2, PA1, PA2.

plastic	Sample Standard Deviation experimental spectrum	synthetic spectrum	Histogram Intersection Similarity	$R_{Ca/H} = I_{CaII}/I_{HI}$ experimental spectrum	synthetic spectrum	Standard Deviation
ABS1	722.085	733.583	0.948	0.662	0.678	0.026
ABS2	632.629	631.971	0.945	0.035	0.036	0.014
PA1	791.998	802.591	0.955	1.073	1.066	0.003
PA2	911.441	941.795	0.934	0.330	0.294	0.037

as the same type of plastic. Meanwhile, the relative intensity ratios of the Ca II spectral lines to the H I spectral line at 656.3 nm $R = I_{CaII}/I_{HI}$ are relatively close between these two types of plastics, requiring the GAN to perform more refined simulation and generation. Therefore, we compared the $R_{Ca/H}$ value of the experimental spectrum with those of the ten synthetic spectra and calculated the standard deviation of the $R_{Ca/H}$ values among the ten synthetic spectra in Table 2. It can be observed that the average value of the experimental spectrum differs negligibly from that of the synthetic spectra, and the standard deviation of the $R_{Ca/H}$ values among the ten synthetic spectra is very small. This indicates that the relative intensity ratios of the spectral lines in the generated spectra remain highly stable, ensuring the quality and physical interpretability of the synthetic spectra.

When only one synthetic spectrum was generated per plastic category, the clustering behavior between the synthetic and experimental spectra could not be clearly demonstrated by PCA clustering. Therefore, t-distributed Stochastic Neighbor Embedding (t-SNE) was employed to analyse both the training and testing sets in the BP neural network. By utilizing the outputs extracted from the hidden layers of the neural network, it was confirmed that the distribution characteristics had been sufficiently learned within the internal representations. As shown in Fig. 8, the effective proximity of the only one synthetic spectrum and the experimental spectrum was observed, indicating that both synthetic and experimental spectra were effectively utilized in model training. Furthermore, the synthetic spectra were found to cluster well with the remaining test samples, demonstrating that the model was properly

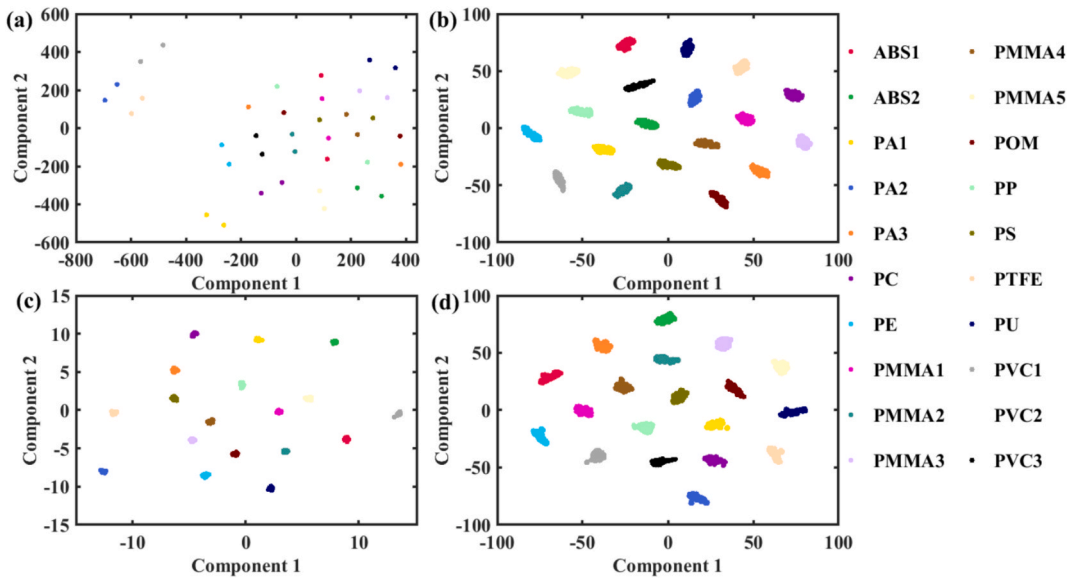


Fig. 8. The t-SNE of (a) 1 synthetic spectrum and 1 experimental spectrum for training (b) 1 synthetic spectrum and 99 experimental spectra for testing (c) 10 synthetic spectra and 1 experimental spectrum for training (d) 10 synthetic spectra and 99 experimental spectra for testing.

trained and capable of effectively distinguishing all 20 plastic categories. Similar trends were observed when multiple synthetic spectra were generated, confirming that the synthetic spectra contributed positively to the model training process.

Using the proposed method, one synthetic spectrum along with one experimental spectrum for each kind of plastic was obtained to form an extended database. To simulate training with only one spectrum, both the input to the generative model and the training spectrum for the classification model were taken as the same spectrum from the 100 collected randomly. After this single spectrum was input into C-GAN for data augmentation, that single spectrum was combined with the synthetic spectra to form the training set for the classification model, while the remaining 99 spectra were used as the test set. Based on this method, the synthetic spectrum and its corresponding experimental spectrum were used together to build a new BP classifier with the same testing spectral data set, where a better classification accuracy at 99.75% was obtained (Fig. 9). Thus, the classification model trained by the extended spectra database was obviously better than only trained by the experimental spectral database.

To avoid the contingencies that come with BP model, repeatability tests were conducted for different methods in one-shot and two-shots. Their average recognition rate and standard deviation were evaluated. In Fig. 10, whether in one-shot or two-shots, the recognition results of training the synthetic spectrum generated by C-GAN with the

experimental spectra together were more accurate and stable than those trained with only with the experimental spectra. By contrast, the CTGAN model cannot generate spectra with only single spectrum for training. Its average recognition rate of the synthetic spectra in the two-shots was also slightly lower than that of C-GAN. This suggests that the internal connections of these spectra can be better obtained. And the synthetic spectra generated by the deep learning tool provide better performance than those experimental spectra with instability caused by manual manipulation. Table 3 illustrates the applications of GANs for spectral-based classification and identification across different materials, all of which significantly improved classification accuracy, demonstrating the feasibility of applying GANs for data augmentation in the field of classification and recognition. And our work has also shown remarkable results.

In order to verify the universality of the research result, 10 groups of spectra were randomly selected from the 100 groups of spectra obtained in the experiment for synthesis. The synthetic spectra were trained together with the synthetic spectra as the training set of the BP training model, and repeated experiments were conducted. The results obtained were in the line chart which was shown in Fig. 11. After data augmentation, the accuracy rate of plastic identification was greatly improved, and the stability of accurate identification was also enhanced.

Besides BP network, SVM and k-NN were also used for evaluating the C-GAN improvement, shown in Fig. 12. Comparisons before and after

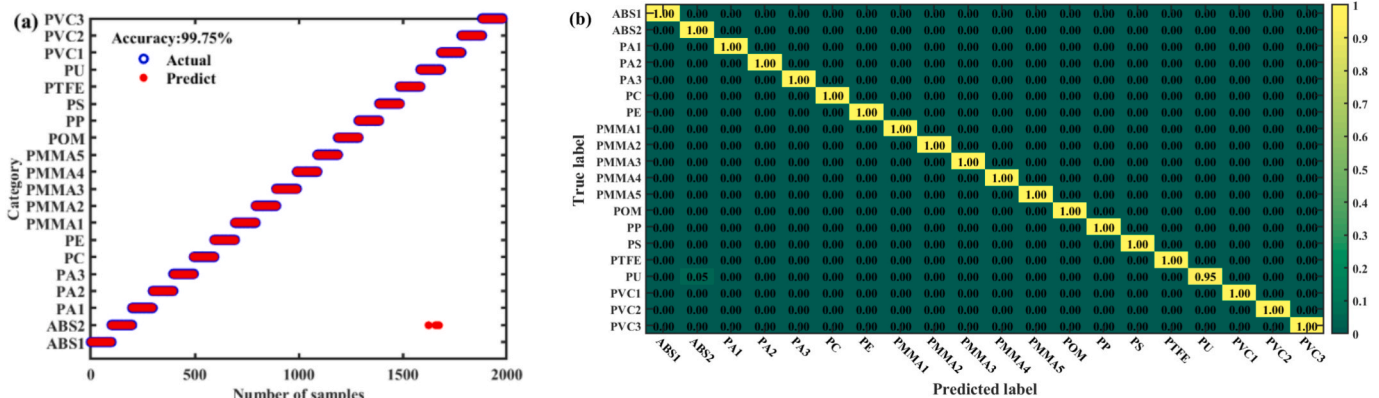


Fig. 9. One-shot experimental spectrum with synthetic spectrum of 20 kinds of plastic classification result (a) and the confusion matrix (b).

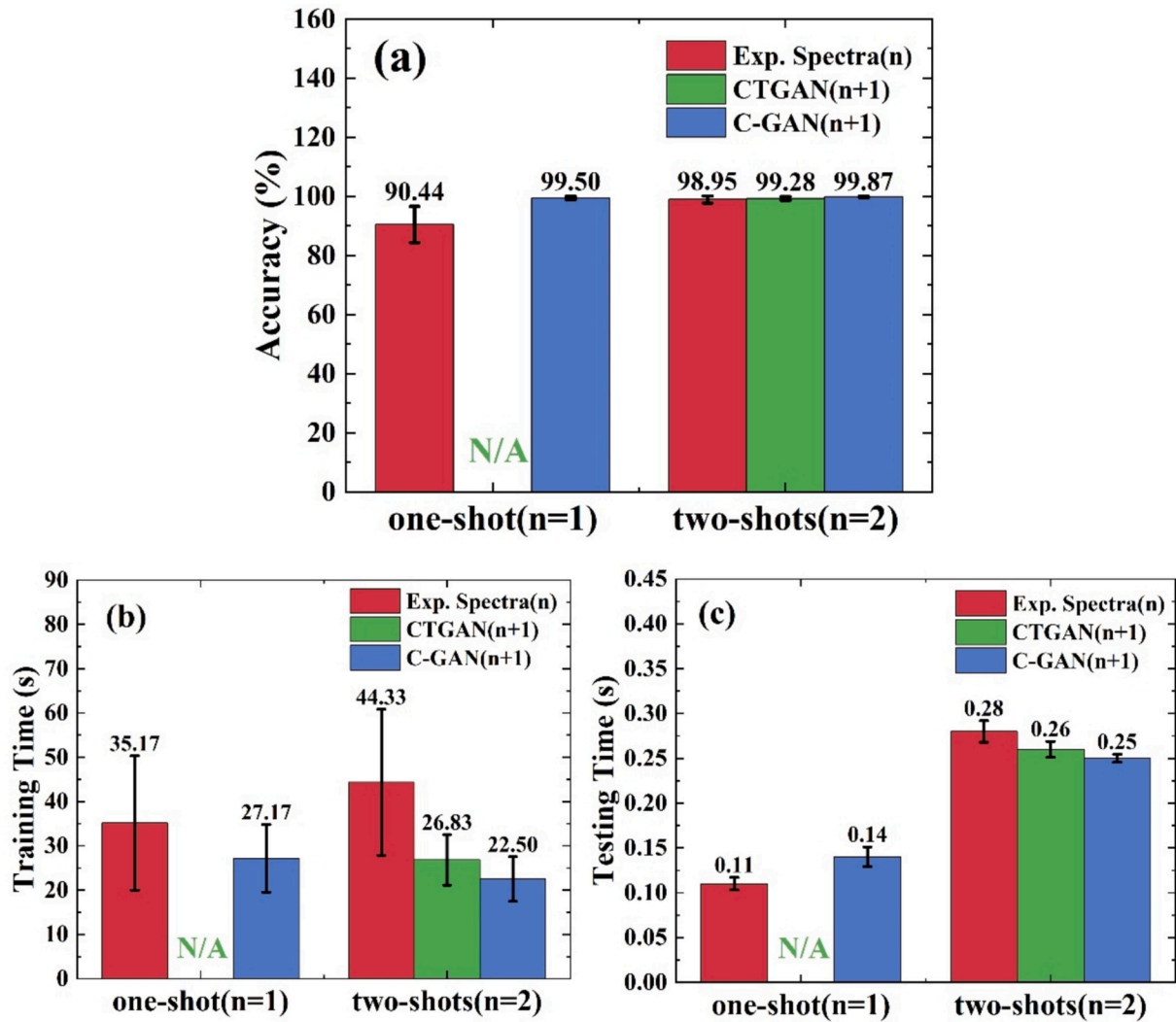


Fig. 10. The accuracy (a), training time (b) and testing time (c) of C-GAN, CTGAN, one-shot ($n = 1$) and two-shots ($n = 2$) experimental spectra (Training set: red-pure n experimental spectra, green- n experimental spectra + 1 synthetic spectrum by CTGAN, blue- n experimental spectra + 1 synthetic spectrum by C-GAN).

Table 3

Applications of GANs in the field of spectral-based classification and identification of different materials.

Technology	Method	Materials	Number of samples – train: test	Classification Accuracy	Author
LIBS	C-GAN	20 plastics	1: 99	99.50%	Our work
	GAN	5 hazardous materials	10: 90	95.33%	Teng et al.[18]
	TCWGAN-GP	14 rock materials	25: 10	97.90%	Jamali et al.[20]
Raman Technology	GAN	3 strains of bacteria	209: 1 (LOOCV)	90%	Du et al.[22]
	PGGAN	5 deep-sea bacterial strains	80%: 20% (5-fold CV)	99.8%	Liu et al.[23]

data augmentation were conducted using training sample sizes of 2, 3, 5, and 10. In the SVM model, learning was not applicable with only one training sample per category, while an accuracy of approximately 92.42% was attained with two training samples (1 experimental spectrum and 1 synthetic spectrum). Similarly, comparative analyses under data augmentation were performed with sample sizes of 3, 5, and 10 in the k-NN model. In the augmented spectra, there was only 1 experimental spectrum, while the rest are synthetic spectra. The recognition accuracy was effectively improved when classification models were trained using multiple synthetic spectra. Sufficiently high recognition accuracy was achieved with an adequate number of synthetic spectra. Of course, it is evident that synthetic spectra generally exhibit more limited

diversity compared to experimental spectra, it's the reason that synthetic spectra underperform relative to experimental spectra. The diversity of experimental spectra primarily arises from random noise and complex perturbations inherent in experimental instruments, in addition to possibly undefined perturbations by experimental operations. In the model, Gaussian white noise is employed to simulate random noise variations, and the approach for simulating intensity variations is also relatively simplistic and confined to the domain of the generative model. Consequently, the relative variation in synthetic spectra is limited, leading to insufficient diversity. As a result, under identical training sample conditions, models trained with augmented synthetic spectra tend to yield lower recognition accuracy than those trained exclusively

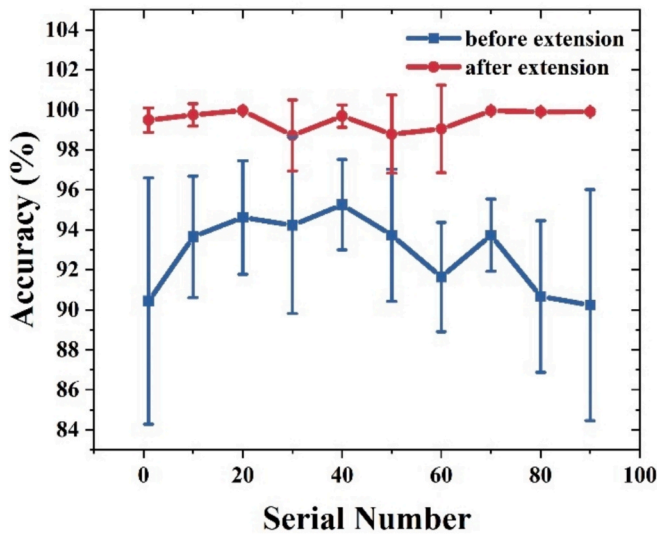


Fig. 11. 10 groups of samples were randomly selected for stability analysis.

with experimental spectra.

5. Conclusion

In this work, we proposed a LIBS data augmentation method to realize one-shot learning for plastic classification. Using LIBS spectra for plastic classification is primarily based on the atomic and molecular lines of the plasma generated from the samples. The wavelength and intensity represent the species and content, respectively. Under different experimental conditions or environments, the wavelengths and intensities of these lines fluctuate slightly. The main variations arise from instrumental fluctuations, such as those in the laser, spectrometer, and noise. Therefore, corresponding fluctuations were incorporated into the model to simulate real conditions. For each kind of plastics, one experimental spectrum was fused with relative noise perturbations, spectral line shifts, and nonlinear intensity scaling, then input into the proposed C-GAN model to obtain a synthetic spectrum. The experimental spectrum and synthetic spectrum were not exactly similar but retained distinct features recognizable by the classifier. For demonstrating the classification accuracy, the data were input into BP, k-NN and SVM networks. The results of the BP network showed that classification accuracy of one experimental spectrum with one synthetic

spectrum addition reached 99.49%, much higher than the one experimental spectrum only (90.44%). The k-NN and SVM also provided similar results, demonstrating that the classification of plastics can achieve the advantages of accurate and efficient machine learning in the actual industrial environment. It can also effectively avoid the disadvantages of machine learning, which requires a large amount of data for training. Moreover, it can reduce the degradation of spectral quality caused by complex industrial environments and simple experimental equipment.

As the number of synthetic spectra used for training increases, it is observed that their performance was slightly inferior to that of experimental spectra. When unseen samples of the same type but from different batches were encountered, differences in additives may exist. Plastic classification using LIBS spectra was determined by atomic and molecular spectral lines of matrix elements (C, H, O, N) in polymers. Additives (Ca, Mg, Zn, etc.) with only small amounts have little influence on the lines of matrix elements. Consequently, although the generated spectra lack sufficient diversity, this lack of diversity does not significantly affect the model's generalization ability on unseen samples. But the decision boundary of a classification model is determined by the distribution of the training samples. When the diversity of the synthetic spectra is poor and their similarity is highly concentrated in certain local regions of the experimental spectra, the classification model may tend to overfit the features of these dense regions while neglecting to learn the marginal features or transition regions of certain classes that the generated samples fail to cover. This can lead to a shift in the decision boundary, preventing the model from correctly learning the distinctions between classes. Additionally, test samples may differ slightly from the training set due to fluctuations in experimental conditions. If the generated samples lack diversity, the model's generalization capability for such out-of-distribution samples will be severely limited, potentially resulting in misclassification of spectra from the same class but acquired under different batches or instruments. So using GAN to generate synthetic sample still has the problem of how to balance the similarity of the samples with the model's classification ability, which still needs to be further studied and overcome in our further work.

CRediT authorship contribution statement

Jiamei Liu: Writing – original draft, Validation, Software, Investigation, Data curation. **Nan Zhao:** Validation, Methodology. **Bin Wang:** Writing – review & editing, Resources. **Laiqun Xia:** Funding acquisition, Data curation. **Qiongxiang Ma:** Writing – review & editing, Methodology, Investigation. **Lei Wang:** Writing – review & editing, Resources.

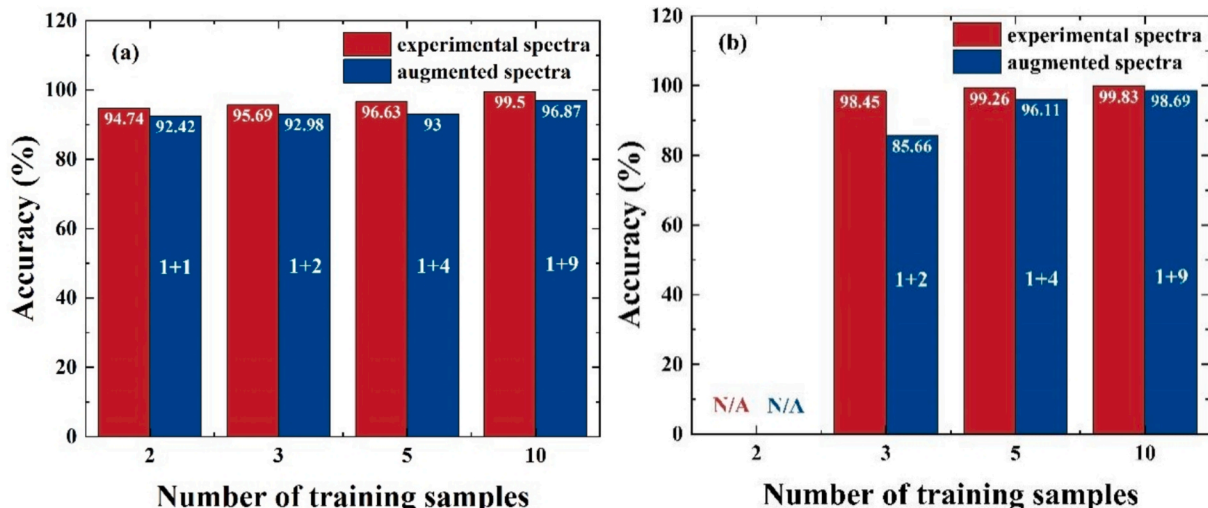


Fig. 12. Comparison of accuracy before and after data augmentation with different numbers of training samples: (a) SVM model; (b) k-NN model.

Jiaming Li: Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Conceptualization. **Qingmao Zhang:** Writing – review & editing, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data underlying the results presented in this paper are not publicly available at this time but maybe obtained from the authors upon reasonable request.

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